Ideation Phase

Define the Problem Statements

Date	28 October 2026
Team ID	Ujjawal Rathod (Individual Capstone)
Project Name	Electric Vehicle Market Performance & Insights
Maximum Marks	2 Marks

Customer Problem Statement Template: A well-articulated customer problem statement focuses on what matters to each stakeholder persona (Emily Johnson, David Chen, and Dr. Sarah Williams) to design an effective analytical solution in Tableau.

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	**Emily Johnson (Market Strategy Director)** — a senior automotive strategist responsible for EV product portfolio planning and budget allocation.	Identify which of the 15 EV categories drive the highest annual sales, track model-year trends (2018–2026), and align driving range with customer ratings.	It takes too long to extract clear insights on category volume, growth stability, and customer satisfaction from raw records.	Data is split across `ev_market_train.csv` (200k rows) and `ev_market_test.csv` (100k rows) with 37 columns and missing values (~1% in sales, ~2% in ratings).	**Frustrated and uncertain** when making high-stakes investment decisions without visual proof.
PS-2	**David Chen (Automotive BI Manager)** — a business intelligence lead tracking global competitor market share, battery adoption, and regional demand.	Benchmark the Top 10 manufacturers across 200 competitors, analyze production across 10 countries, and spot regional product positioning opportunities.	Standard Top-N filters distort market share percentages, and flat regional totals (~330M per region) obscure category-specific demand.	The EV market is highly fragmented (top manufacturer Ampere Works holds ~0.59% share), requiring FIXED LOD calculations and a Region × Category heatmap.	**Stressed and concerned** about misinterpreting competitor dominance or missing regional expansion pockets.

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PS-3	**Dr. Sarah Williams (Sustainable Mobility Researcher)** — an EV researcher evaluating battery chemistries, energy efficiency, and range trends.	Compare how battery types (NMC, LFP, NCA, Sodium-Ion, Solid-State) affect driving range (km), efficiency (Wh/km), and customer satisfaction over time.	Default zoomed-in chart axes make minor random fluctuations (~319–329 km) look like dramatic technology shifts, hiding the true drivers of range.	Real-world driving range in this dataset is dictated by vehicle category (550 km Luxury Sedan vs. 105 km Scooter), while battery types average ~324 km across all years.	**Skeptical and cautious** about drawing flawed sustainability conclusions without properly scaled baselines.